

Harsh Gupta

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Education

IIT Kharagpur	B.Tech, Industrial Engineering & M.Tech, Financial Engineering	8.39	2027
National High School	Higher Secondary Education (CBSE)	91.2%	2022
National High School	Secondary Education (CBSE)	91%	2020

Work Experience

Member of Technical Staff Intern | DevRev May 2026 – Sep 2026

- Architected trajectory-level evaluation for agent actions and tool calls, surfacing 35% more failure modes than output-only evaluation.
- Engineered a custom-scorer framework enabling domain-specific agent evaluations; adopted by 42% of active organizations.
- Reduced CircleCI runtime 55%, from 56 to 25 minutes, by importing prebuilt Go module layers instead of resolving dependencies.

Software Engineer Intern | Sidecar Aug 2025 – Mar 2026

- Configured Caddy as a reverse proxy behind Cloudflare, routing multi-domain traffic across 10+ internal services on Azure VMs.
- Established a centralized observability platform with Prometheus, Grafana, Loki, and Alertmanager, unifying Azure resource telemetry and service logs into dashboards and alerts; deployed an on-call triage bot to surface root-cause hypotheses during incidents.
- Hardened Azure access with a Tailscale private mesh, securing development services without exposing internal endpoints.

Full Stack Engineer Intern | Chainwide Jul 2024 – Dec 2024

- Engineered async ETL across Salesforce, Confluence, and Notion APIs with backoff, idempotency, and vector-store normalization.
- Created a Slack Bolt RAG assistant routing queries across 3 Sales, Service, and Marketing agents with isolated vector namespaces.
- Built RAG observability dashboards tracking 5+ quality and latency metrics, improving answer quality 40% via LLM-as-judge tuning.

Machine Learning Engineer Intern | Practworks Dec 2023 – Apr 2024

- Modeled 700+ learning topics and prerequisite edges as a Directed Acyclic Graph, applying topological sort to generate learning paths.
- Generated 48,000 synthetic learner trajectories from behavioral data of 1,200 students, augmenting sparse data for cold-start training.
- Trained a Random Forest next-topic recommender on learner interactions; achieved 84% accuracy, 22% above a rule-based baseline.

Software Engineer Intern | Ylytic Sep 2023 – Dec 2023

- Delivered 5 influencer-analytics features by integrating Instagram and YouTube APIs into a unified analytics layer.
- Optimized the influencer-data pipeline by 65%, from 60 to 21 seconds, using asyncio concurrency and batch database writes.
- Implemented Pydantic schema validation and structured error logging, capturing 99.9% of processing failures with diagnostic context.

Entrepreneurial Experience

Founder & Software Developer | Vanguard AI | github.com/dudegladiator/AI-Inventory-Manager-3.0 Mar 2024 – Feb 2025

- Won a Rs. 500K grant for a multilingual AI inventory platform combining barcode management and voice-driven retail workflows.
- Designed a voice-first inventory assistant supporting 3 regional languages for intent-aware add, remove, and price-update commands.
- Created a tool-calling AI analytics assistant accessing inventory data and operational tools for on-demand retailer insights.

Competitions

Opensoft '26: Financial Trading Simulation | Silver Medal | github.com/dudegladiator/Financial-Trading-Simulation Mar 2026 – Apr 2026

- Built a C++ limit-order book supporting price-time FIFO matching, AVL-tree price levels, and real-time delta-depth updates.
- Sandboxed custom Python strategies in isolated Docker (256MB, 0.5 CPU), competing against built-in bots on a shared exchange.
- Devised an event-driven market-data pipeline using Redis Streams, Pub/Sub, and PostgreSQL batch writers for trades and OHLCV.

Projects

Personal Hedge Fund Jan 2025 – Apr 2025

- Assembled ETL pipelines aggregating real-time market, fundamental, and news data from 5+ sources into a normalized datastore.
- Orchestrated 3 tool-augmented research agents for fundamental, technical, and sentiment analysis across live and historical data.
- Implemented a scheduled portfolio monitor aggregating P&L and exposure, triggering alerts for price moves and allocation drift.

Skills and Expertise

Languages/Backend: Python, C++, Go, TypeScript, SQL; FastAPI, Django, React, REST, WebSockets, gRPC, Flask

Data/Infra: PostgreSQL, Redis, MongoDB, Neo4j(Graph DB), Git, Docker, Linux, Azure, GCP, CI/CD, Datadog, Grafana, Figma, Excel

Coursework: Data Structures & Algorithms, Software Engineering, Probability & Statistics, Big Data Analysis, Statistical Learning

Leadership and Extracurriculars

Quant Club, IIT Kharagpur May 2023 – Present

- Conducted quantitative research and authored a white paper on algorithmic trading, documenting methodology and findings.
- Launched Summer of Quant, a structured upskilling program on Python, statistics, and algorithmic trading for 4,000+ members.
- Organized the WorldQuant BRAIN Consultant Challenge with industry professionals, engaging 250+ participants.

Achievements and Awards

Silver and Bronze medals at Opensoft GC 2026 and 2025 for trading simulation and Mindflix	2025, 2026
Placed second in Optima's Hacktank Competition for a video-proctoring system	2025
Selected for dual-degree M.Tech in Financial Engineering through merit-based departmental upgrade	2024
Led Azad Hall's Illumination team to Gold at IIT Kharagpur's 2024 Diwali	2024
Won first prize in the nationwide Bhashini Sprint Competition for an AI voice-controlled inventory platform	2024